Sed × precip suppression: 625 m results (12 h)

7 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz. All 72 runs at 625 m: Nd 20–1000 × sed ×0.5/1/2 × precip suppression on/off × 3D turbulence on/off. Each is 12 h, run in restart segments (checked bit-for-bit), with means over hours 10–12. The 625 m column is added next to the 40 km–1.25 km runs from the 12 h note. The main figures use 3D turbulence off.

Findings

1. **The runs are steady.** $|d\mathrm{LWP}/dt| \leq 4.5\,\mathrm{g\,m^{-2}\,h^{-1}}$ over hours 10–12 in every 625 m run.
2. **A clear inverted V.** With sed×1 and suppression on, LWP goes $52 \to 98 \to 120 \to 122 \to 119 \to 115\,\mathrm{g\,m^{-2}}$ for Nd 20 → 1000, and peaks near Nd 200. $S_{200\to1000}$ is $-0.016/-0.038/-0.064$ for sed ×0.5/1/2 (Fig. **??**).
3. **The falling branch does not need precip suppression.** With suppression off, $S_{200\to1000}$ is still negative ($-0.007/-0.027/-0.033$), and LWP falls from 109 to $99\,\mathrm{g\,m^{-2}}$ between Nd 50 and 1000 while w_e rises from 6.3 to $7.3\,\mathrm{mm\,s^{-1}}$. The high-Nd decline is entrainment-driven, and sedimentation strengthens it.
4. **The low-Nd branch is steepest yet.** With suppression on, $S_{20\to50} = 0.60/0.76/1.00$ for sed ×0.5/1/2. With suppression off it is $-0.06/0.08/0.42$, which comes from deck break-up.
5. **The trend with resolution continues from 1.25 km to 625 m.** LWP is lower (about 100–135 vs. 120–$180\,\mathrm{g\,m^{-2}}$), w_e is higher (up to $7.9\,\mathrm{mm\,s^{-1}}$), and peak cloud cover is lower (≈0.94). The Nd response has the same shape at 2.5 km, 1.25 km and 625 m, with the inverted V somewhat sharper at 625 m.
6. **3D turbulence makes almost no difference at 625 m** (Fig. **??**). LWP changes by −1.0 to $+1.5\,\mathrm{g\,m^{-2}}$ (<1.5 %), cloud cover by less than 0.007 and LWP SD by less than $2\,\mathrm{g\,m^{-2}}$. The one consistent signal is w_e, which is 0.02–$0.06\,\mathrm{mm\,s^{-1}}$ lower (<1 %). This matches the 6 h ACI-study result. Any effect should appear at 312 m.

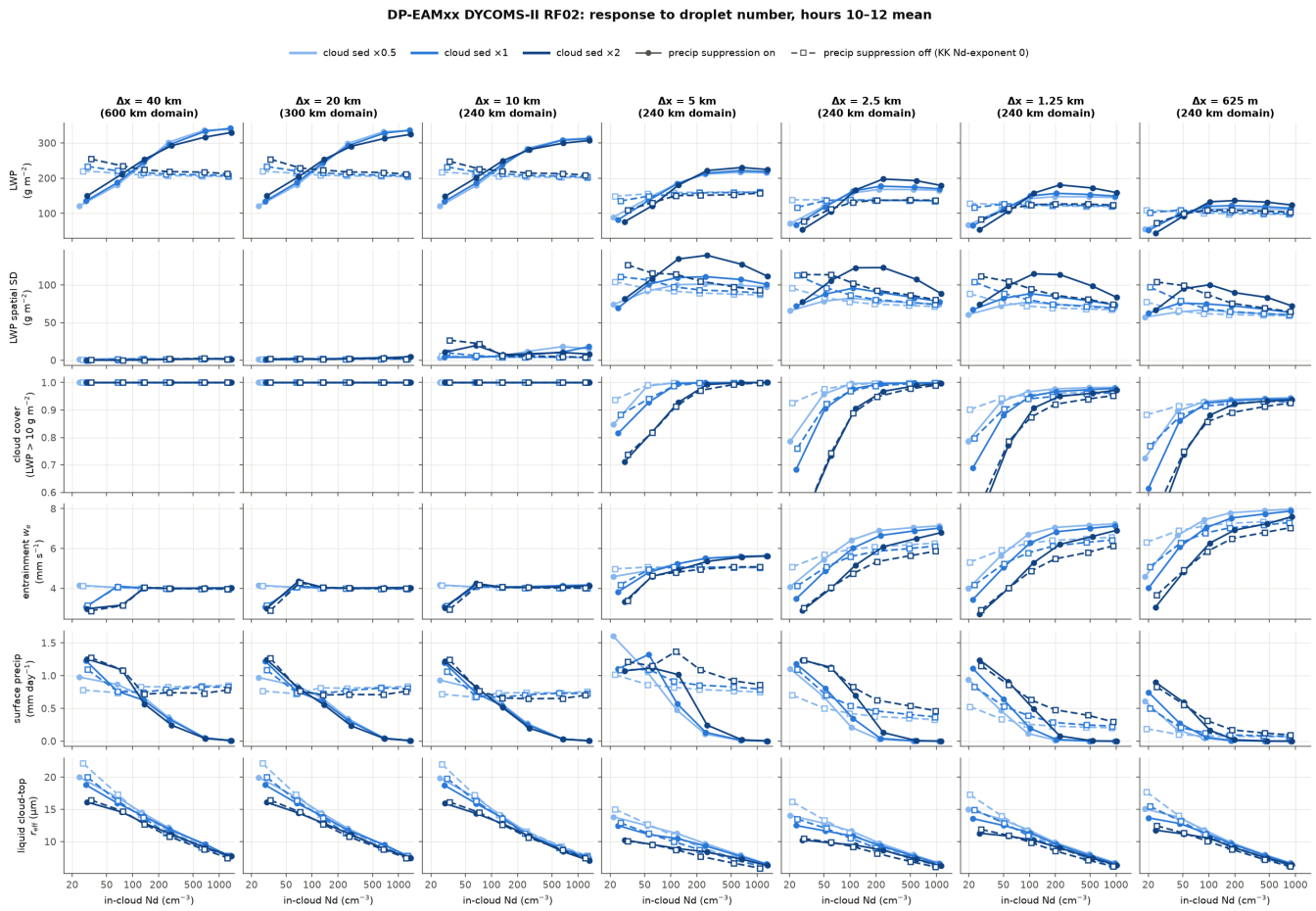


Figure 1: Hour 10–12 means against diagnosed Nd, 40 km to 625 m (3D turbulence off).

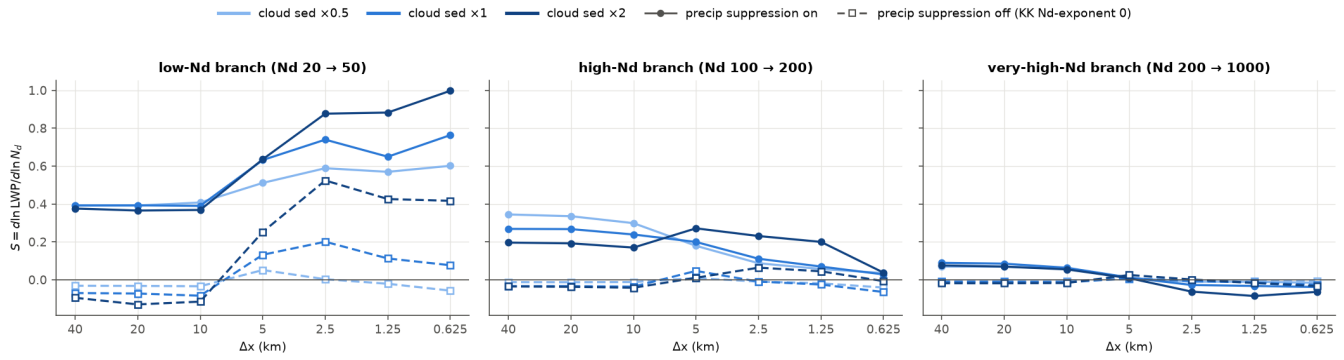


Figure 2: Susceptibility on three Nd branches against resolution, hours 10–12.

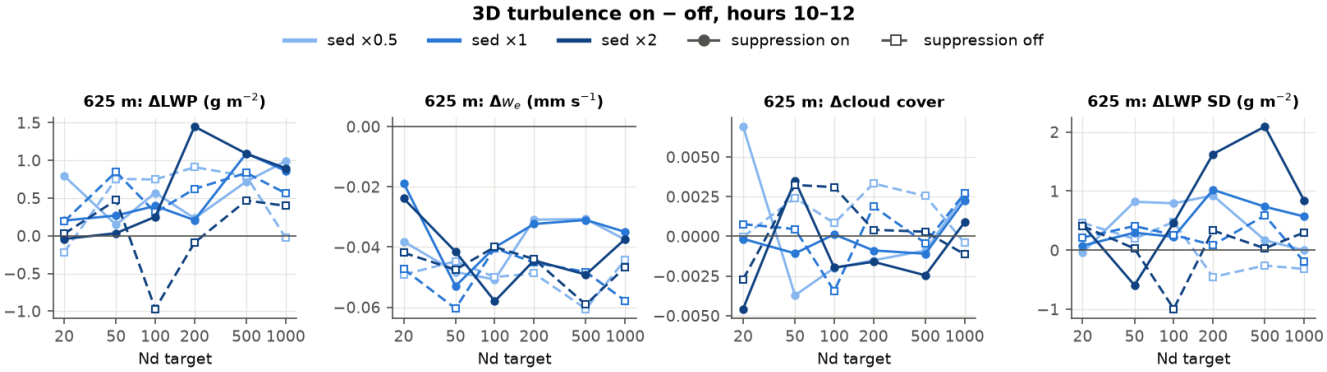


Figure 3: 625 m: 3D turbulence on minus off, for every Nd, sed and suppression setting.

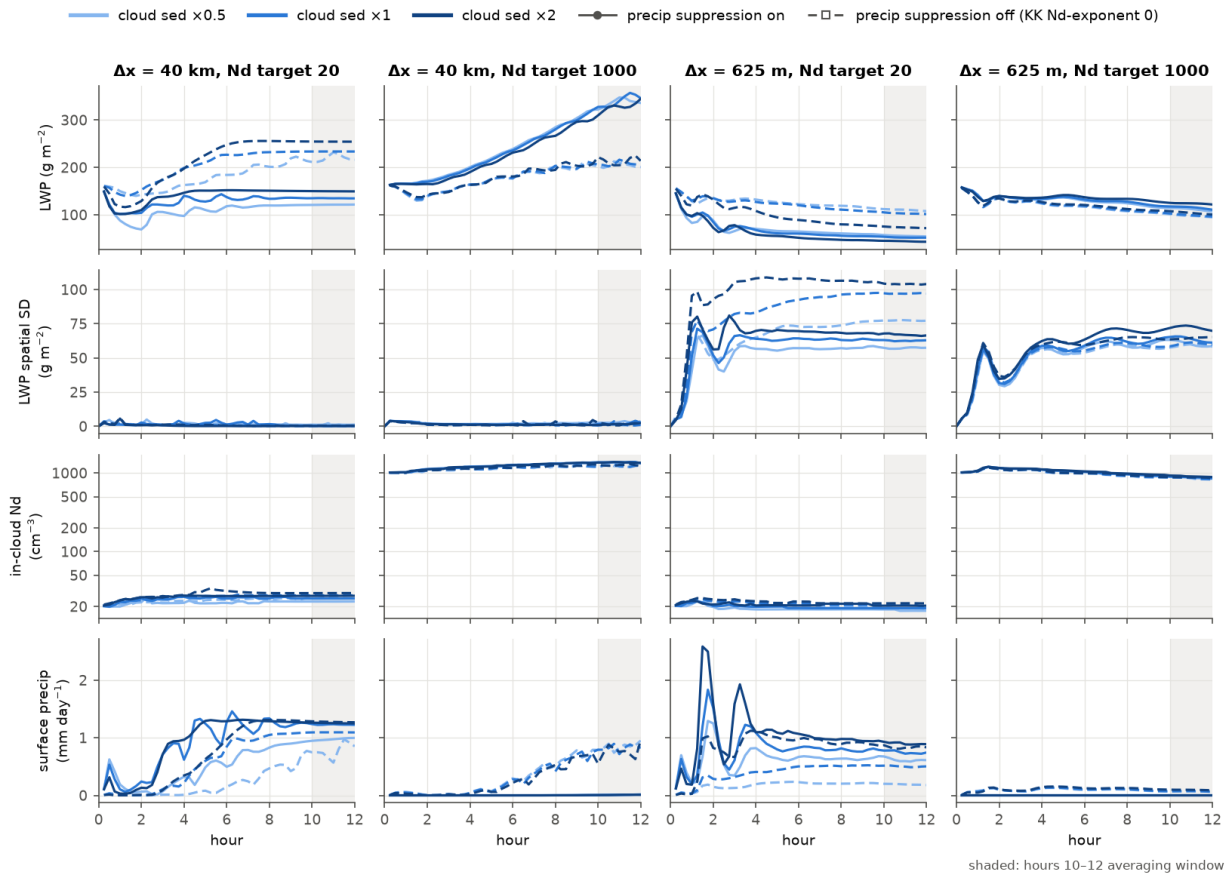


Figure 4: Time series at 40 km and 625 m, Nd 20 and 1000.

Data: 625 m runs are in `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/sedps_625m_*`, with case names ending in `_t3don` for 3D turbulence on. Combined 12 h analysis (40 km–625 m): `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/analysis_with625m/`. It contains `summary_h10-12.csv` and `timeseries_15min.csv` (both with a `turb3d` column), `susceptibility.csv`, `turb3d_on_minus_off.csv` and the figures. 312 m is in progress.